

PM v1: Supplemental Methods and Evidence

Review additions prepared on 14 September 2026

These are insertable paragraphs and tables for the available v1 draft, not a replacement for the latest manuscript. Archived statistics and new diagnostic results are identified separately. The Chinese supplement report records the full execution history and limitations.

Development uses controlled synthetic states, and external evaluation uses EvoEmo virtual users. The resource advantage is relative to the two specified structured-resource baselines; the current evidence does not establish a benefit over Context Only or a cost-matched fixed action.

Rule Source Selector. The query concatenates the current-session summary, visible recent messages, and current user message. The rule hashes word unigrams and bigrams into 64 nonnegative, L2-normalized dimensions and compares the query with each available source’s catalog fingerprint by cosine similarity. It selects up to two sources above the frozen thresholds $\theta_{MP} = \theta_{MS} = \theta_{ME} = 0$. Strategy RAG is enabled when the maximum lexical cosine similarity to a Strategy Bank card is at least 0.08. The rule uses catalog fingerprints; the final stable PM uses dialogue and inventory metadata without these fingerprints. On the 204 scored external states, the rule selects MS+ME+RS in every case; this is a property of this sample, not a constant-policy definition of the rule.

Inventory variants and development units. Each of the 192 synthetic source states is expanded into nine inventory variants: no memory, MP, MS, high-relevance ME, low-relevance ME, MP+MS, MP+high-ME, MS+low-ME, and MP+MS+high-ME. High and low ME are selected by lexical similarity to the same current context, with memory identifiers breaking ties; both ME variants expose one item. Variant labels are construction-only and are not model inputs. The resulting 1,728 cards contain 192, 768, 576, and 192 cards with 2, 4, 8, and 16 legal actions, respectively, yielding 11,136 state-action responses. The 16 users are grouped into five folds, keeping all variants of a source state together. User identities are held out; the 36 current-user utterances and 12 semantic families are reused across folds. Thus, this is a controlled learnability diagnostic rather than a test of unseen language families.

Development score and prediction scales. The development response-preference score is distinct from the external 1–5 Emotional Support rating. Within each card, pair labels are coded as $y \in \{-1, 0, 1\}$ and projected onto action score differences using $\hat{z} = (B^\top B + 0.10I)^{-1} B^\top y$, where each row of B contrasts two actions. After centering $\hat{z}$, the diagnostic score of an action is its mean $\sigma(\hat{z}_a - \hat{z}_b)$ against the other legal actions. The deployed logistic ranker is trained on non-tie feature differences with mirrored orientations and equal total weight per card. Its sigmoid-transformed action score is an ordering score, not an independently calibrated probability of successful support. The five ridge heads predict normalized annotation targets and are clipped to $[0, 1]$; they are not clinical risk estimates.

Legality, feasibility, and fallback. Legal actions respect source availability. Feasible actions additionally satisfy predicted-risk thresholds. The frozen setting is $\epsilon = 0$, $\tau_{mis} = 0.35$, and $\tau_{omit} = \tau_{str} = 1$. Since predictions are bounded by one, the latter two thresholds do not exclude actions. The highest predicted response score is selected within the feasible set; exact ties are resolved by estimated cost, memory decision quality, strategy decision quality, response score, and action identifier. The cost proxy sums estimated top- k memory tokens, 260 tokens if RS is enabled, and a penalty of 24 per source invocation. It is not measured end-to-end latency or total generator input. Confirmatory inference stops with an error if no action is feasible. A least-violation fallback is available only in explicitly enabled development searches, and settings that use it are excluded from final validation selection.

External memory and fixed inputs. MP contains separate profile fields; MS contains historical session summaries. Each external ME item concatenates the seeker messages from one past session in temporal order; ME is not constructed from gold event timelines. The generator receives the most recent eight prior messages and the current seeker message, with an empty current-session summary in this adapter. The

Table 1: Existing original V4 comparisons with Context Only. Differences are PM minus Context Only over 204 states; intervals use scenario clusters. Values are extracted from the saved statistical summary. The original per-state score file was unavailable in this checkout for independent recomputation. No simultaneous coverage is claimed.

Metric	Difference	95% CI
Emotional Support	-0.088	[-0.167, -0.015]
Personalization	-0.098	[-0.157, -0.039]
Memory Appropriateness	-0.054	[-0.093, -0.015]
Factual Grounding	-0.010	[-0.034, +0.015]
Temporal Consistency	-0.010	[-0.029, +0.010]
Non-Intrusiveness	-0.025	[-0.054, +0.005]

204 scored states comprise 34 scenarios, three fixed seeker seeds, and turns 3 and 8, shared across policies. These seeker seeds are not independent PM training seeds.

Comparison with Context Only. The original six-candidate GPT-4o evaluation does not show an external response benefit for structured resource injection over Context Only. PM minus Context Only is -0.088 for Emotional Support and -0.098 for Personalization. The corresponding previously saved scenario-cluster 95% intervals are [-0.167, -0.015] and [-0.157, -0.039]. These are unadjusted per-comparison intervals. Thus, lower input relative to Rule and Fixed Structured should not be interpreted as establishing that learned routing is preferable to omitting long-term resources.

Which decision stages affect the selected action? An offline replay exactly reproduces PM and Rule actions on all 204 external states. All states have 16 legal actions. No highest-score tie occurs, so cost tie-breaking changes no decision. Misuse filtering removes candidates in 200 states but never changes the final choice relative to unconstrained response-score maximization. Omission and strategy thresholds exclude no actions, and no fallback is used. On this external sample, the chosen action is therefore identical to the response ranker’s unconstrained argmax; resource savings cannot be credited to an empirically active cost tie-break. In a separate replay of 1,728 out-of-fold development predictions using the final consensus thresholds, misuse filtering changes 97 choices, while cost tie-breaking again changes none. This replay characterizes decisions and does not measure causal changes in response quality.

Strategy preference and transfer. Among 5,568 development response comparisons that hold the memory configuration fixed and swap RS/R0, the saved dual-order labels favor RS in 2,673 cases and R0 in 1,313, with 1,582 ties. Out-of-fold predictions rank RS higher in 4,147 of these action pairs. On external states, RS is ranked higher in 1,536/1,632 matched-memory action pairs, and the selected RS rate is 192/204. This directly describes the response-ranking behavior; the later strategy-quality tie-break is not needed to explain these external choices. The separately scored ESConv diagnostic uses GPT-4o and favors R0 more often. These differences motivate a transfer/calibration limitation, but do not identify judge family as the cause.

Cost-matched fixed routing. ME+R0 was chosen using development validation under a mean cost proxy no greater than PM’s, not by searching external response scores. Matching is an average-budget criterion rather than exact per-turn equality. The archived external diagnostic reports mean inputs of 1,294.6 for PM and 1,283.8 for ME+R0, and PM-minus-ME+R0 Emotional Support of -0.172 under a separate two-candidate GPT-4o protocol. An archived 40-state, 80-call forced-swap probe reports a two-order-average difference of -0.150 with a unit-bootstrap interval of [-0.250, -0.050]. These absolute scores are not merged into the six-candidate table. The archived report and available code were recovered; their referenced raw score files were not available for independent recomputation here. The evidence supports reporting a competitive simple fixed policy, not a demonstrated external advantage for learned routing.

Resource savings and fixed-input seeds in the rerun. On the 204 rerun states with Filter OFF, PM uses 20.70% fewer generator input tokens than Rule and 22.03% fewer than Fixed Structured. The paired token differences are -332.76 (user-cluster 95% CI [-385.09, -283.35]) and -360.14 ([-412.50, -310.45]), respectively. The input-saving direction is consistent across all three fixed seeker seeds. These intervals describe the 204 scored states and should not be attached to the original 1,020-turn population means.

Seed-specific quality differences vary, and candidate position is confounded with seed and turn in the original rerun ordering; the seed check therefore does not establish quality robustness.

Candidate-order sensitivity in the rerun. We selected 48 states without consulting response outcomes: eight per seeker-seed/turn stratum, covering all 18 users and 34 scenarios. We rescored the same six completed replies in their original and exactly reversed orders with GPT-4.1-mini-2025-04-14, retaining all other prompt fields and serialization order. All 96 quality requests completed and were checked against their uploaded requests and returned response identifiers. For Filter OFF, PM-minus-Rule Overall changes from +0.104 to -0.042; PM-minus-Fixed changes from +0.146 to 0.000. Their two-order averages are +0.031 (user-cluster 95% CI [-0.061, +0.170]) and +0.073 ([-0.051, +0.225]), respectively. The point estimates are sensitive to candidate order. The averaged differences remain small and uncertain; this is not a passed equivalence test or validation of the original GPT-4o table’s candidate order.

Exploratory demand for historical information. A response-blind judge inspection of all 204 rerun states used only authorized history and the actual generator-visible context window, excluding current-topic evaluator annotations. Its first outputs labeled 169 states as potentially helped by additional history and 35 as not needing additional history; none was labeled as requiring an unavailable historical fact to answer an explicit recall request. Twelve separate synthetic annotation controls passed. However, 35 first outputs failed verbatim historical-citation checks, and 25 remained unresolved after the frozen retry limit. All original labels and failures are retained. These are exploratory LLM annotations, not a validated recall benchmark or evidence that historical information is unnecessary. The existing separate omission scorer assesses missing useful authorized context, including evaluator-only current-topic information; it should not be relabeled as a pure cross-session memory-recall measure.

The evidence establishes a deployable pre-retrieval source allocator, learnable allocation signal in controlled synthetic development, and reduced input relative to specified high-resource structured policies. It does not yet establish that this learned PM is preferable to simple low-cost configurations, or that the resource reduction preserves performance on a validated history-required task. Such a claim would require an independently specified probe with verified needed facts, retrievable evidence, and a scorer shown to distinguish omission.